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Co-exposure of chromium or cadmium and a low concentration of amoxicillin are responsible to emerge amoxicillin resistant *Staphylococcus aureus*

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Background: Heavy metals and antimicrobials co-exist in many environmental settings. The co-exposure of heavy metals and antimicrobials can drive emergence of antimicrobial resistant (AMR) *Enterobacteriaceae*. We hypothesized that co-exposure to heavy metals and a low concentration of antibiotic might alter antimicrobial susceptibility patterns, which facilitate emergence of AMR *Staphylococcus aureus*.
Methods: The growth kinetics of antimicrobial susceptible *S. aureus* was carried out in the presence of chromium or cadmium salt and a low concentration of antibiotics. Subsequently, the antimicrobial susceptibility pattern was determined by the Kirby-Bauer disc diffusion method. Moreover, the mRNA copy number was determined by reverse transcription polymerase chain reaction.
Results: The antimicrobial susceptibility profile revealed that the zone of inhibition (ZOI) for ampicillin, amoxicillin, ciprofloxacin and doxycycline was significantly decreased in chromium pre-exposed *S. aureus* compared to unexposed bacteria, whereas cadmium pre-exposed bacteria only showed significant decreased in ZOI for amoxicillin. Moreover, the MIC of amoxicillin for *S. aureus* was increased by 8-fold in chromium and 32-fold in cadmium when bacteria were co-exposed with low concentrations of amoxicillin. The mRNA expression of *femX, mepA* and norA also significantly increased in *S. aureus* after exposure to chromium and a low concentration of amoxicillin.
Conclusion: Cultivation of *S. aureus* at the minimum levels of chromium or cadmium and a low concentration of amoxicillin increased the inhibitory concentration of amoxicillin through inducing bacterial efflux pumps and antibiotic resistant genes. However, it is warranted to assess the whole transcriptome to find out all responsible factors behind this de novo amoxicillin resistant *S. aureus.*

1. Introduction

Antibiotic resistance is one of the major global problems and threatens the usefulness of nearly all antimicrobials used to alleviate microbial infections [1]. Several studies have proved that antimicrobial agents other than antibiotics have the capability to cause antibiotic resistance through a co-selection process [2,3]. In addition to antibiotic resistance, heavy metal contamination is another severe ecological problem [4,5]. Antibiotic and heavy metals co-exist in the environment such as in the gastrointestinal tract, animal manure, and poultry farming sites [6–9]. For instance, arsenic and antibiotics are widely used in poultry farms as growth promotion and disease control agents (DOI: 10.1126/science.336.6083.795). Thus, the gut microbiota of domestic animals is getting exposed to both antibiotics and heavy metals. Subsequently, fertilizers made from manure and sewage sludge containing both antibiotics and heavy metals mix with agricultural soil that eventually leaches into the water [3,10]. As a result, there is a high possibility that humans are exposed to these heavy metals through contaminated food chain or several other means [4,11]. The combined effect of both antimicrobials and heavy metal

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residues may exert strong influences on the emergence of AMR both in human and environmental settings. However, the evidence of combination effects of AMR is not clearly defined. Several studies have reported that the elevated levels of heavy metals are responsible to emerge de novo antibiotic resistant bacteria [3,7,12]. For example, the growth of LSJC7 (e.g., an *Enterobacteriaceae*) has significantly increased in arsenate and tetracycline co-exposure milieu [9]. We therefore examined the effect of AMR in a single *S. aureus* strain upon exposure of both heavy metals and a low concentration of clinically relevant antibiotics in this study.

The antimicrobial sensitivity profile and growth kinetics of *S. aureus* was determined in the presence of chromium or cadmium salts. This study has revealed that growth of *S. aureus* with a low concentration of chromium or cadmium and amoxicillin was significantly increased the minimum inhibitory concentration (MIC) of amoxicillin, which may facilitate to emerge the de novo amoxicillin resistant *S. aureus*.

2. Methods and materials

2.1. Place of study

This study was carried out at the Infection and Immunity Laboratory in the Department of Biochemistry and Molecular Biology, University of Dhaka, Bangladesh.

2.2. Collection of Staphylococcus aureus

In this study, the *Staphylococcus aureus* was isolated from processed raw meat (meat ball) from a local restaurant by Food Microbiology Laboratory, Laboratory Sciences and Services Division, International Center for Diarrhoeal Disease Research, Bangladesh (ICDDR,B).

2.3. Dose-response growth kinetics in presence of chromium (Cr^{6+}) or cadmium (Cd^{2+}) salt

Single colonies of *Staphylococcus aureus* isolates from Mueller Hinton agar (MHA) culture plate were grown in 0.5 mM, 1 mM, 3 mM, 5 mM, 10 mM, 50 mM, 100 mM chromium salt or grown in 0.005 mM, 0.01 mM, 0.025 mM, 0.05 mM, 0.075 mM, 0.1 mM, 0.3 mM, 0.5 mM, 0.75 mM cadmium salt in Tryptone Soya broth (TSB). The optical density (O.D) was measured at 600 nm using UV-Vis spectrophotometer (Thermo-Scientific).

2.4. Pre-screening of antibiotic susceptibility pattern

A total of 10 μL (e.g., 10^8 cfu/mL) inoculum of *S. aureus* was grown in 0.0 mM, 0.5 mM, 1.0 mM chromium or 0.0 mM, 0.05 mM, 0.1 mM cadmium salt containing TSB media at 37°C, 180 rpm for 12 h. Subsequently, 100 μL of each chromium or cadmium pre-exposed bacterial suspensions were spread on Mueller-Hinton agar plates. The standard antibiotic discs, Azithromycin (15 μg), Chloramphenicol (30 μg), Ampicillin (10 μg), Amoxicillin (10 μg), Erythromycin (15 μg), Doxycycline (30 μg), Ciprofloxacin (5 μg), Cephradine (30 μg), Clindamycin (2 μg), Methicillin (5 μg) and Vancomycin (30 μ) were impregnated on Mueller-Hinton agar plates, which were pre-inoculated with chromium or cadmium pre-exposed *S. aureus*. After 20 h of incubation at 37°C, the antibiotic susceptibility profile was determined by Kirby-Bauer disc diffusion method. The zone of inhibition (ZOI) was annotated following the Clinical Laboratory Standard Institute (CLSI, USA) guideline, 2017 for *S. aureus* (Supplementary Table S1).

2.5. Determination of bacterial growth kinetics in heavy metal and antibiotic co-exposure setting

To determine the co-exposure effect of chromium or cadmium and amoxicillin on *S. aureus* growth, 10 μL (equivalent to 10^8 cfu/mL) of bacterial suspension was inoculated in 0.5 mM chromium or 0.025 mM cadmium or with a 0.06 μg/mL of amoxicillin (e.g., the minimal inhibitory concentration) containing TSB media and incubated in a shaker incubator at 37°C with 180 rpm. The optical density of was measured at 600 nm as previously described (9).

2.6. Determination of minimum inhibitory concentration of amoxicillin for co-exposed S. aureus

S. aureus was grown in TSB with or without 0.5 mM chromium and a low concentration of amoxicillin (0.06 μg/mL) for 48 h at 12 h re-inoculated (e.g., subculture) interval. Similarly, *S. aureus* was also grown in cadmium (0.025 mM) and a low concentration of amoxicillin (0.06 μg/mL) co-exposure setting. Similarly, the reference strain of *S. aureus* (ATCC #6538) was grown in TSB supplemented with/without chromium or cadmium salt. After 24 to 72 h, 10 μL of bacterial suspension (e.g., 10^5 cfu) was inoculated in 0.06 to 4.0 μg/mL of amoxicillin containing agar plates and incubated for 20 h at 37°C. The minimum inhibitory concentration (MIC) for each condition was recorded following the guideline of Agar dilution method [13]. Moreover, the amoxicillin susceptibility levels were also determined for the above-mentioned co-exposed bacteria following the CLSI guideline as described in Section 2.4 (CLSI, guideline 2017, M100, 27th edition).

2.7. Determination of efflux activity of chromium or cadmium pre-exposed reference strain of S. aureus

S. aureus ATCC #6538 was pre-exposed with a low concentration of chromium or cadmium for 72 h with 12-h subculture intervals. Then 100 μL (e.g., 10^7 cfu) bacterial suspension was inoculated in PBS supplemented with 2 μg/mL ethidium bromide (EtBr) and 0.25% glucose for a period of 30 min. After replacing the buffer with EtBr-free PBS with or without glucose, the efflux activity was assessed by fluorescence plate reader (Agilent BioTek Synergy LX) with a semi-automated fluorometric method, which performed at 530 nm excitation and 600 nm emission spectrums for 30 min at 5-min intervals.

2.8. Extraction of RNA from S. aureus *isolates and cDNA synthesis*

S. aureus is lysozyme-resistant due to the presence of modified peptidoglycan layer [14]; hence, to recover maximum yield of RNA, the simple phenol method is the most effective method for bacterial cell lysis. After phenol treatment, the Monarch Total RNA Miniprep Kit (New England Biolab) was used for bacterial RNA isolation. To check the quality of RNA, agarose gel electrophoresis was performed. Next, cDNA synthesis of purified RNA was performed using ProtoScript II First Stand cDNA synthesis Kit [15].

2.9. Quantification of mRNA level of norA, mepA and femX using reverse transcription polymerase chain reaction (RT-qPCR)

To quantify the expression level of norA, mepA and femX, the *S. aureus* was grown for 48 h in a chromium and amoxicillin co-exposure setting as described in Section 2.6. The cDNA copy number of each gene was determined by reverse transcription polymerase chain reaction (RT-qPCR) using the above-mentioned prepared cDNA (Table 4). The RT-qPCR was performed using the following PCR primers (Tables 1 and 2) and PCR-master mix (New

Table 1
Primer sequence, size, and product size.

Name of Gene	Primer sequence	Length (bp)	Product size (bp)
femX	5'GCGAAGAATCGCTGTAGGTC3'	20 (forward)	193
	5'TGCATACGCTTTCTCAGCTT3'	20 (reverse)	
norA	5'TGGCCACAATTTTCGGTAT3'	20 (forward)	182
	5'CACCAATCCCTGGTCCTAAA3'	20 (reverse)	
mepA	5'TGCTGCTGCTCTGTTCTTTA3'	20 (forward)	198
	5'GCGAAGTTTCCATAATGTGC3'	20 (reverse)	
GAPDH	5'TGACACTATGCAAGGTCGTTTCAC3'	24 (forward)	180
	5'TCAGAACCGTCTAACTCTTGGTGG3'	24 (reverse)	

Table 2
List of reagents for RT-qPCR.

Name of reagent	Volume
Template: cDNA	1.0 µL
SYBR-Green master mix	10 µL
1 µM primer Forward (25 nM)	1.0 µL
1 µM primer Forward (25 nM)	1.0 µL
Nuclease free water	7.0 µL
Total volume	20 µL

England Biolabs). The relative gene expression was normalized through comparing with housekeeping GAPDH, using the comparative threshold cycle as previously described [16,17].

2.10. Statistical analysis

All statistical analyses were performed using GraphPad Prism version 6.0. The analysis of variance (one-way ANOVA) and Sidak multiple-comparison test were performed. Data were expressed as mean ± SEM (standard error of mean). Values of $P < 0.05$ were considered statistically significant.

3. Results

3.1. Isolation of Staphylococcus aureus and culture

Staphylococcus aureus (*S. aureus*) was isolated from processed raw meat (meat ball) from a local restaurant by Food Microbiology Laboratory, Laboratory Sciences and Services Division, International Centre for Diarrhoeal Disease Research, Bangladesh. The standard reference strain of *S. aureus* (ATCC #6538) was collected from department of Microbiology, University of Alabama at Birmingham, USA. *S. aureus* is a Gram-positive coccus and facultative anaerobic microorganism. It is a non-fastidious bacterium and grow well in Tryptone Soya medium. In this media, at mid-log phase (O.D. 0.5 at 600 nm) the average viable count was approximately 10^8 cfu/mL, which was counted by Miles-Mishra serial dilution method.

3.2. Growth kinetics of S. aureus in the presence of chromium salt

S. aureus was grown in TSB supplemented with or without chromium (Cr^{6+}) salt. The growth curve (Fig. 1A) demonstrated that *S. aureus* was viable in up to 3 mM Cr^{6+} salt, but the bacterial growth was inhibited at 5 mM or higher concentrations chromium containing medium. Therefore, 0.5 to 3.0 mM chromium salt was used in exposure assays as the concentration allows growth of this strain of *S. aureus*. Similarly, we found that *S. aureus* was viable in up to 0.1 mM cadmium (Cd^{2+}) salt solution, but the bacterial growth was inhibited at ≥0.3 mM cadmium salt solution (Fig. 1B). Therefore, a range of 0.005 to 0.1 mM cadmium salt solution was used during exposure assessment of *S. aureus*.

3.3. Antibiotic sensitivity pattern of S. aureus after exposure to chromium/cadmium

The antibiotic susceptibility profile of *S. aureus* following exposure to a range of 0.5 to 1.0 mM Cr^{6+} revealed that the zone of inhibition (ZOI) for amoxicillin was significantly decreased from 36 mm to 27 mm in *S. aureus* isolates. Similarly, the ZOI for ciprofloxacin was significantly decreased in chromium exposed *S. aureus* isolates. Furthermore, the ZOI was significantly decreased in *S. aureus* isolates against amoxicillin from 37 mm to 30 mm after exposure to a range of 0.05 to 0.1 mM Cd^{2+} for 48 h. There was no significant change in the ZOI for other antimicrobials tested (Table 3).

3.4. Co-exposure effects of chromium or cadmium and a low concentration of amoxicillin on S. aureus growth

S. aureus was grown in Tryptone soya broth, supplemented with or without 0.5 mM chromium salt or 0.06 µg/mL amoxicillin, or both chromium and amoxicillin. The growth curve demonstrated that the chromium pre-exposed *S. aureus* isolates growth was comparable with chromium un-exposed bacterial isolates (Fig. 2A). However, after treatment with amoxicillin, the growth of chromium pre-exposed or exposed *S. aureus* was significantly

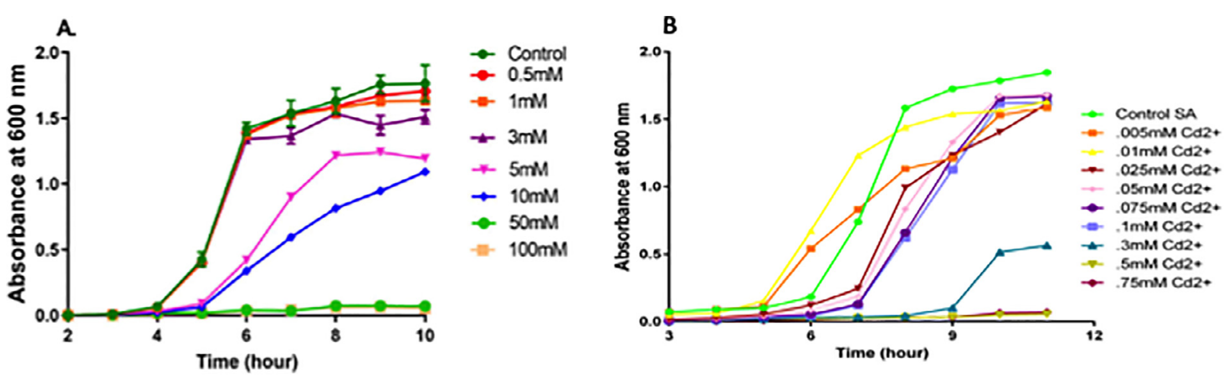

Fig. 1. Assessment of chromium and cadmium salt concentration tolerable to *S. aureus* growth. (A) Growth curve of *S. aureus* with different concentrations of chromium salt in TSB; (B) Growth curve of *S. aureus* in different concentrations of cadmium salt in TSB. The data show mean ± SEM and n = 3.

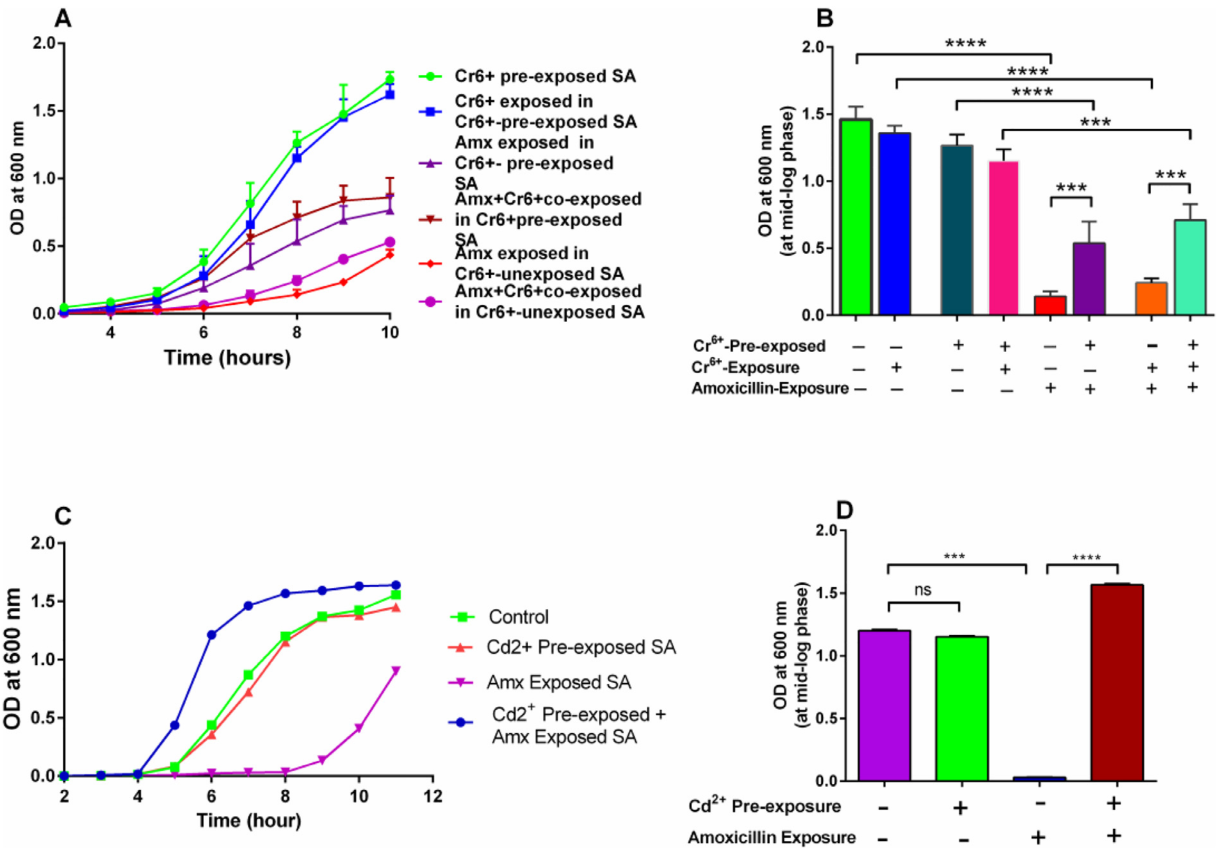

Fig. 2. Co-exposure effect of chromium and amoxicillin or cadmium and amoxicillin on *S. aureus* growth. (A) Growth curve of *S. aureus* in presence of chromium salt and amoxicillin. (B) Growth rate of *S. aureus* at mid-log phase (at 8 h post-exposure). (C) Growth curve of *S. aureus* in presence of cadmium salt and amoxicillin. (D) Growth rate of *S. aureus* at mid-log phase (at 8 h post-exposure). Amx, amoxicillin; SA, *S. aureus*. The data show mean ± SEM and P *** < 0.01, ****$P < 0.001$, n = 3.

Table 3
Zone of inhibition of *S. aureus* against clinically important antibiotics following exposure to chromium or cadmium.

Name of antibiotic	Antibiotic susceptibility of *S. aureus* with and without exposures to chromium salt			Antibiotic susceptibility of *S. aureus* with and without exposures to cadmium salt		
	Level of Chromium	ZOI (mm)	P value	Level of Cadmium	ZOI (mm)	P value
Amoxicillin	0mM	36	0.05	0.0	37	0.01
	0.5mM	30		0.05mM 0.1mM	33	
	1.0mM	27			30	
Ciprofloxacin	0mM	35	0.05	0.0	32	NS
	0.5mM	34		0.05mM 0.1mM	30	
	1.0mM	30			28	
Azithromycin	0mM	29	NS	0.0	27	NS
	0.5mM	29		0.05mM 0.1mM	26	
	1.0mM	26			26	
Chloramphenicol	0mM	28	NS	0.0	32	NS
	0.5mM	27		0.05mM 0.1mM	31	
	1.0mM	26			31	
Ampicillin	0mM	36	0.05	0.0	37	NS
	0.5mM	36		0.05mM 0.1mM	35	
	1.0mM	33			35	
Erythromycin	0mM	30	NS	0.0	32	NS
	0.5mM	30		0.05mM 0.1mM	32	
	1.0mM	30			32	
Clindamycin	0mM	31	NS	0.0	34	NS
	0.5mM	31		0.05mM 0.1mM	33	
	1.0mM	29			32	
Doxycycline	0mM	31	0.05	0.0	34	NS
	0.5mM	31		0.05mM 0.1mM	33	
	1.0mM	29			31	
Cephradine	0mM	32	NS	0.0	32	NS
	0.5mM	31		0.05mM 0.1mM	31	
	1.0mM	30			30	

NS, non-significance.
Data were analyzed, the analysis of variance (one-way ANOVA) and Sidak multiple comparison tests were performed.

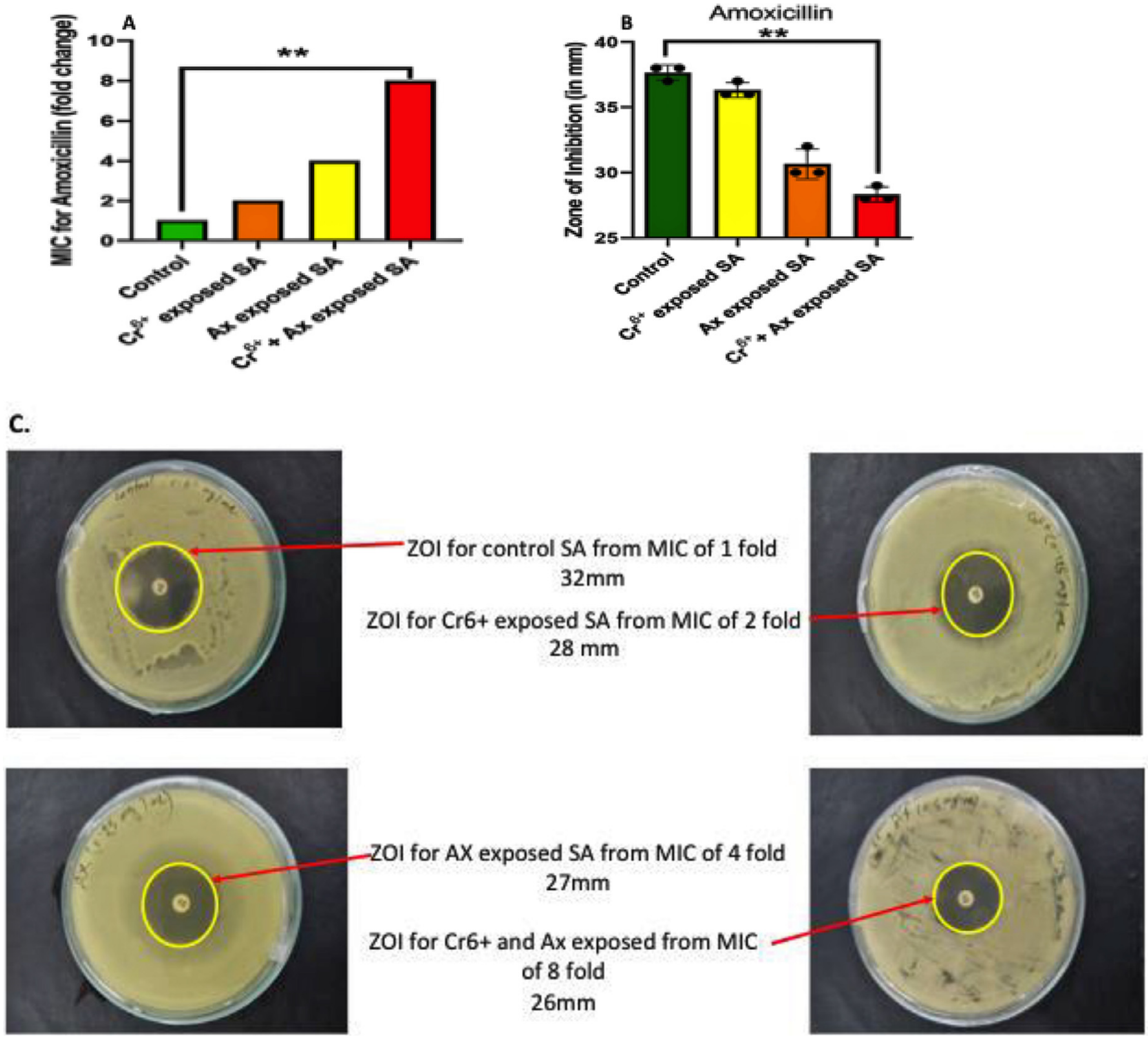

Fig. 3. Susceptibility patterns of chromium and amoxicillin pre-exposed *S. aureus* against amoxicillin: (A) shows the fold change of MIC for amoxicillin. Briefly, *S. aureus* was pre-exposed with a lower dose of chromium and amoxicillin for 48 hours and then 10 μL (e.g., 10^5 cfu) of 10 times diluted bacterial suspension was loaded on TSA plates containing different levels of Amoxicillin and incubated for 20 h. The control media contains no antibiotic. (B) and (C) show the bar chart of ZOI and representative antimicrobial susceptibility test images for amoxicillin. Briefly, *S. aureus* was pre-exposed with a lower dose of chromium and amoxicillin for 48 h and then 100 μL (e.g., 10^7 cfu) bacterial suspension was spread on each plate and incubated for 20 h. Data are shown as mean ± SEM and statistical analysis was performed with one-way ANOVA and Sidak's post hoc test, $^{**}P < 0.01$, n = 3. SA, *S. aureus.*

increased compared to the chromium unexposed control. Quantitively, amoxicillin or chromium treatment alone decreased bacterial growth by 52.3% compared to control, whereas the bacterial growth rate was enhanced up to 77.3% when exposed to a combination of 0.5 mM chromium salt and 0.06 μg/mL amoxicillin. Overall, *S. aureus* growth kinetics demonstrated that chromium pre-exposure decreased the inhibitory effect of amoxicillin, suggesting that chromate enhanced amoxicillin resistance. Moreover, the mid-log phase growth kinetics demonstrated that bacteria exposed to amoxicillin showed decreased growth compared to isolates with exposure but co-exposure to both chromium and amoxicillin significantly increased bacterial growth (Fig. 2B). Furthermore, bacterial growth rate was low after chromium or amoxicillin exposure, but the growth rate was restored by exposure to chromium and amoxicillin in combination. These observations, suggest that the combination of amoxicillin and chromium exposure reverses the inhibitory effect of either agent alone.

Furthermore, the growth curve demonstrated single exposure with amoxicillin decreased bacterial growth by 52.3% compared to unexposed *S. aureus*, whereas the growth rate was enhanced by 82.6% by exposure to cadmium combined with a low concentration of amoxicillin (Fig. 2C). The growth kinetics at the mid-log phase showed that there were no significant changes in terms of growth rate between unexposed and cadmium pre-exposed *S. aureus* isolates, whereas the growth rate of unexposed was sharply reduced after amoxicillin treatment. The inhibitory growth rate was retrieved after co-exposure with both cadmium and amoxicillin (Fig. 2D). Overall, *S. aureus* growth kinetics suggest that the growth pattern of *S. aureus* was altered following exposure to both heavy metal and antibiotic compared to single exposure with either heavy metal or antibiotic.

3.5. MIC of amoxicillin following exposure to heavy metals in S. aureus

The agar dilution method was performed to determine the minimum inhibitory concentration (MIC) of amoxicillin for *S. aureus* under chromium and amoxicillin co-exposure setting. First, chromium or antibiotic unexposed *S. aureus* isolates, the MHA plates showed only 2 to 3 cfu at each spot in the MHA plate sup-

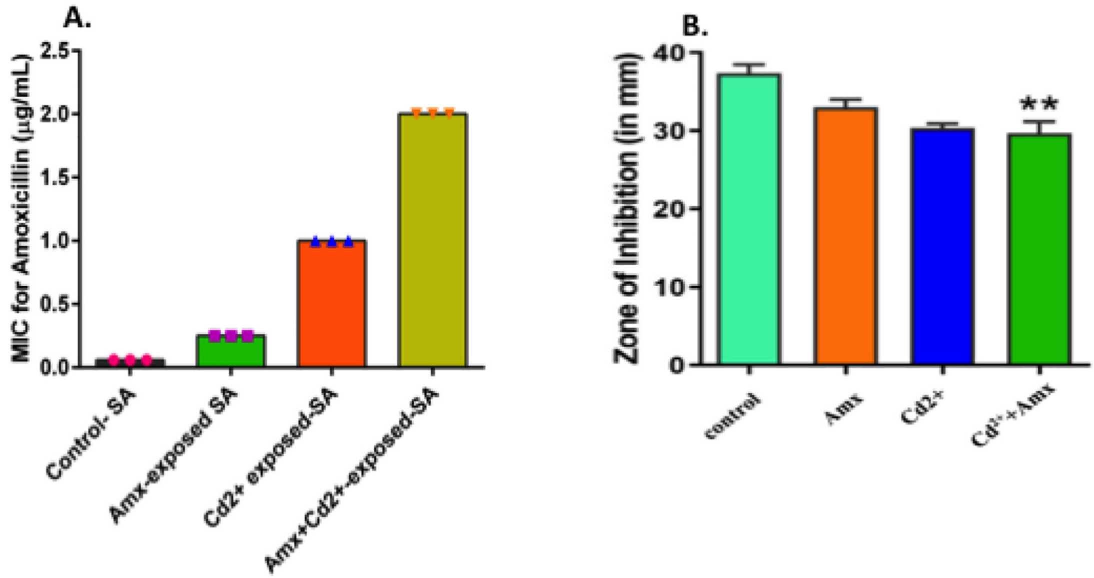

Fig. 4. Amoxicillin susceptibility patterns of cadmium and amoxicillin pre-exposed *S. aureus*: (A) shows the MIC for amoxicillin. Briefly, *S. aureus* was pre-exposed with a low concentration of cadmium and amoxicillin for 48 h. 10^5 cfu bacteria was inoculated on TSA plates containing different levels of amoxicillin and incubated for 20 h. (B) shows the ZOI for Amoxicillin. Briefly, *S. aureus* was pre-exposed with a low concentration of cadmium and amoxicillin for 48 h and then 10^7 cfu bacterial suspension was spread on each plate and incubated at 37°C for 20 h. Data are shown as mean ± SEM and statistical analysis was performed with one-way ANOVA and Sidak's post hoc test, $^{**}P < 0.01$, n = 3.

plement with 0.06 µg/mL of amoxicillin. Whereas no detectable cfu were observed for unexposed *S. aureus* in the MHA plates supplemented with ≥0.125 µg/mL amoxicillin (Supplementary Fig. S1). Therefore, the MIC of amoxicillin was 0.06 µg/mL for unexposed *S. aureus.* However, the MIC of amoxicillin increased to 0.125 µg/mL for chromium pre-exposed bacteria, 0.25 µg/mL for amoxicillin pre-exposed bacteria and 0.5 µg/mL for pre-exposure with chromium and amoxicillin. Overall, the MIC of amoxicillin for *S. aureus* was increased by 8-fold by pre-exposure to a low concentration of chromium and amoxicillin, whereas chromium pre-exposure alone increased the MIC by 2-fold and amoxicillin pre-exposure alone increased the MIC by 4-fold compared to control (Fig. 3A).

3.6. Antimicrobial susceptibility patterns of chromium pre-exposed S. aureus

S. aureus was grown in liquid broth media supplemented with a concentration 0.06 µg/mL, 0.125 µg/mL, 0.25 µg/mL, 0.5 µg/mL after pre-exposure to under chromium or amoxicillin alone or both in combination. From the measurements of zone of inhibition (ZOI) for the amoxicillin antibiotic disc, baseline data shows that *S. aureus* was resistant to amoxicillin 10 µg (e.g., zone of inhibition was 28 mm) under each condition of pre-exposure. The ZOI for untreated bacteria was 32 mm, whereas for chromium or amoxicillin alone or both treatments combined this value was 28 mm, 27 mm and 26 mm, respectively (Fig. 3B-C). Overall, the susceptibility patterns of amoxicillin have demonstrated that co-exposure of chromium and a low concentration amoxicillin significantly decreased the size of ZOI for amoxicillin.

On the other hand, the co-exposure effect of cadmium and a low concentration of amoxicillin increased the MIC of amoxicillin for *S. aureus* by 32-fold compared to unexposed control, whereas the MIC of amoxicillin was increased by 4-fold and 16-fold, respectively, in amoxicillin and cadmium pre-exposure settings, compared to unexposed control bacteria (Fig. 4A).

3.7. Antibiotic susceptibility pattern of cadmium pre-exposed S. aureus

S. aureus was further grown in TSB medium, using the bacteria with altered MIC after incubation with or without pre-exposure to cadmium, amoxicillin or the combination of both. These bacterial suspensions were spread on Mueller-Hinton Agar Media (Supplementary Fig. S1). From the measurements of zone of inhibition for the amoxicillin antibiotic disc, baseline data shows that *S. aureus* was resistant to amoxicillin at 10 µg (e.g., zone of inhibition was 28 mm) under all conditions. The zone of inhibition was significantly decreased in the various co-exposure conditions. For example, the ZOI for unexposed bacteria was 32 mm, for amoxicillin or cadmium or at co-exposure condition the ZOI was 27 mm, 26 mm and 25 mm, respectively (Figure 4B and Supplementary Fig. S2).

3.8. Amoxicillin susceptibility patterns and efflux activity of chromium or cadmium pre-exposed reference strain of S. aureus

The agar dilution method was performed to determine the MIC of amoxicillin against the reference strain of *S. aureus* FDA 209 (ATCC #6538) upon exposure with chromium or cadmium. Upon exposure to chromium or cadmium the MIC of amoxicillin against *S. aureus* subspecies FDA 209 was increased to 0.125 µg/mL at 48 h and 0.25 µg/mL at 72 h. Overall, the MIC of amoxicillin was increased by 2-fold at 48 h and by 4-fold at 72 h chromium or cadmium post-exposure setting compared to unexposed control (Fig. 5A and Supplementary Fig. S3). Furthermore, the efflux assay demonstrated that the efflux of ethidium bromide noticeably increased in chromium or cadmium pre-exposed *S. aureus* compared to unexposed control (Fig. 5B).

3.9. Expression patterns of genes encoding efflux pumps and femX gene in S. aureus

As demonstrated in antimicrobial susceptibility patterns, co-exposure of amoxicillin and heavy metals like chromium and cadmium alter amoxicillin susceptibility and promote emergence of

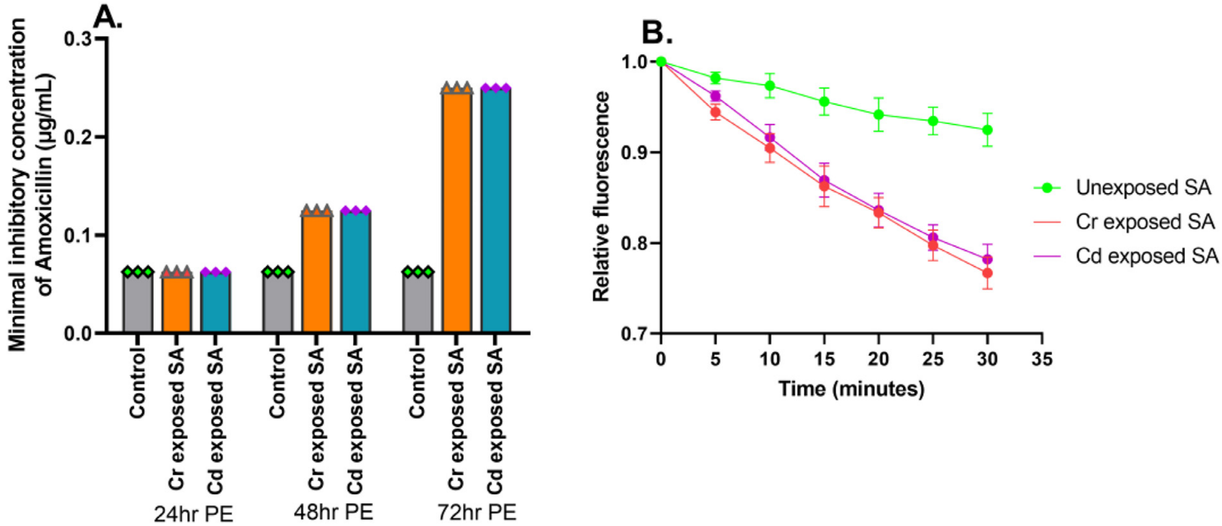

Fig. 5. Amoxicillin susceptibility patterns and efflux activity of chromium or cadmium pre-exposed *S. aureus* subspecies aureus FDA 209: (A) shows the MIC for amoxicillin. Briefly, *S. aureus* ATCC #6538 was pre-exposed with a low concentration of chromium or cadmium for 24 to 72 h, and then 10 µL bacterial suspension was inoculated on TSA plates containing different levels of amoxicillin. (B) shows the fluorescence-based efflux of ethidium bromide (EtBr). Briefly, *S. aureus* ATCC #6538 was pre-exposed with a lower concentration of chromium or cadmium for 72 h and then 100 µL bacterial suspension was inoculated in PBS with 2 µg/mL EtBr and 0.25% glucose for 30 min. The EtBr efflux was assessed by the semi-automated fluorometric method. PE, pre-exposed, SA, *S. aureus*.

amoxicillin resistant *S. aureus*. We hypothesized that this alteration of antimicrobial susceptibility might be associated with the changes of expression of resistance genes or efflux pumps of *S. aureus*. To address this hypothesis, the expression levels of factor C (femX), which catalyses the incorporation of the first glycine of the pentaglycine interpeptide bridge, efflux pumps mepA and norA of *S. aureus* were assessed by RT-qPCR at RNA level.

The overall quality of RNA preparation was assessed by electrophoresis on a denaturing agarose gel. Figure 6A shows the 5s RNA smear, which assures the quality and yield of RNA. Wells with light band denoted disintegrated RNA, which were further extracted. Moreover, to assure the target genes at DNA level and the quality of RT-qPCR gel run was performed. Figure 6B shows the bands for femX, mepA, norA and house-keeping gene GAPDH after RT-qPCR. As the product size of RT-qPCR for the above genes was very close, the bands of all genes were found in the same alignment after Agarose gel electrophoresis. Figure 6C showed that all the targeted genes (femX, mepA, norA, GAPDH) were amplified following the corresponding C**t** value (number of threshold cycle) for control, amoxicillin, chromium and both amoxicillin-chromium pre-exposure condition. Figure 6D shows that the resistance-related factors *femX, mepA,* and *norA* all were significantly upregulated in chromium and chromium-amoxicillin combination pre-exposed *S. aureus* compared with control. Quantitatively, the mRNA expression of femX, mepA and NorA were increased by 25-fold, 19-fold and 17-fold in chromium pre-exposed *S. aureus* compared to control. Whereas, the co-exposure condition of chromium-amoxicillin, the expression of femX, mepA and norA increased by 7-fold, 10-fold and 8-fold, respectively. In contrast, the expression pattern of these genes was not altered in amoxicillin pre-exposed *S. aureus*. Overall, these findings indicate that heavy metals pre-exposed or heavy metal-antibiotic in combination pre-exposure condition has up-regulated the expression of drug-resistant genes as such norA, mepA and femX (Figure 7).

4. Discussion

S. aureus a significant healthcare superbug, which is frightening as its resistance to antibiotics is dramatically increasing day by day. Notably, the irrational use of antibiotics in livestock and heavy metals in leather industries untreated effluents [18] increased the likelihood of bacterial co-exposure in water bodies. Moreover, combined contamination of heavy metals and antibiotics contribute to the emergence of microbial antibiotic or sometimes multidrug resistance [2].

In this study, we hypothesized that heavy metals may alter the level of antibiotic susceptibility and facilitate the emergence of multidrug-resistant *S. aureus*. To address this hypothesis, the antimicrobial susceptibility profiles were assessed of naturally isolated *Staphylococcus* from processed raw meat and culturally grown *Staphylococcus* with a minimum tolerable level of chromium or cadmium. This investigation shows that *S. aureus* can tolerate up to 3 mM $K_2Cr_2O_7$ and 0.5 mM $CdCl_2.H_2O$ salts. Previously, a study reported that the minimum inhibition concentrations (MICs) varied from strain to strain of *Staphylococcus* [19]. For example, the MIC of $CuSO_4.5H_2O$, $Cd(NO_3)_2.4H_2O$, $NaAsO_2$, and $ZnSO_4.7H_2O$ was 2, 0.25, 1, 0.25 mM for *S. haemolyticus* BB02312, for *S. aureus* RN4220 was 4, 0.015, 4, 0.015 mM and for *S. haemolyticus* NW19A the MIC was 8, 0.4, 4, 8 mM, respectively [19]. whereas the MICs of Cu^{2+}, Zn^{2+}, Cd^{2+}, $Cr_2O_7{}^{2-}$, and Ag^{+}, respectively, were 16, 10, 2.5, 1.6 and 0.25 mM for LSJC7, a member of *Enterobacteriaceae* [9].

Moreover, the antimicrobial susceptibility profile of this study revealed that minimal inhibitory concentrations of chromium or cadmium and amoxicillin pre-exposure are able to drive amoxicillin resistant *S. aureus*. The alteration of antimicrobial susceptibility in chromium and cadmium salt pre-exposed bacteria might be responsible for emerging antibiotic resistance superbugs in environmental reservoirs. The current observations are strongly supported by a recent systemic review, where they showed fourteen published research articles reported the co-occurrence of heavy metal and antibiotic resistance [20]. Under selective pressure, microorganisms also acquire antimicrobial resistance by cross-resistance when the route of heavy metals and antimicrobial agents accessed to their target bacteria are similar but not the same, allowing them act in combination [2]. Evidence of cross-resistance can be found in studies with heavy-metal-contaminated environments demonstrating potential microbial adaptation to the environmental selective pressure by acquisition of resistance [21]. The most common form of cross-resistance results from the micro-

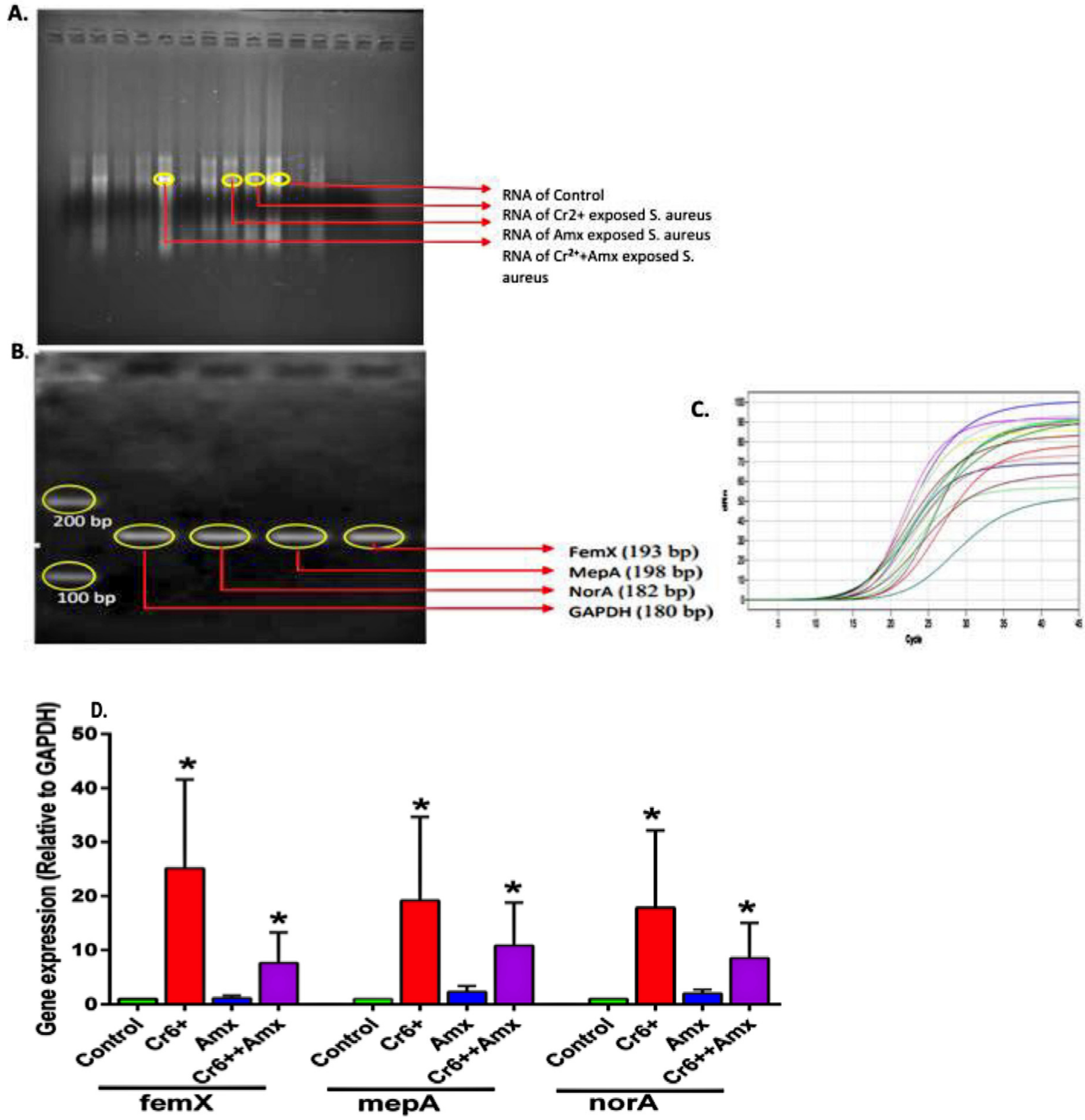

Fig. 6. Relative gene expression in chromium and/or amoxicillin pre-exposed *S. aureus*. (A) Band of 5s RNA in Agarose gel electrophoresis visualized under gel doc. (B) PCR band for product size. (C) Gene amplification. (D) The bar chart for the relative expression of norA, mepA and femx. Data are presented as fold changes on the basis of calculation of gene expression using relative quantification, RQ = $2^{-\Delta\Delta Ct}$. Data are shown as Mean ± SEM and statistical analysis was performed with unpaired *t* test (two-tailed), nonparametric, Mann-Whitney *t* test. ($n = 3$, asterisk* represents upregulation, $^{*}P < 0.05$). Amx, amoxicillin.

bial utilization of an efflux pump, a cellular membrane transport protein [21].

To assess efflux pumps and resistance gene expression at RNA level, the current study adopted RT-qPCR. The RT-qPCR data demonstrated that the expression of femX, mepA and norA have significantly increased after pre-exposure to chromium and a low concentration of amoxicillin in *S. aureus*, as compared to the unexposed group. These findings were supported by previous studies with different bacteria, heavy metals and antibiotic co-exposure settings [9,22,23]. Moreover, a previous study also demonstrated that following metabolic stress, cell transporters especially efflux pumps, acting antiporters, remove heavy metals and antibiotics from cells in a cross-resistance mechanism, which is known as cross-regulation [2].

Furthermore, one of the main mechanisms for microorganisms' acquisition of antibiotic resistance under heavy metals selective pressure is co-resistance. This mechanism is favoured by evolution of multiple resistance via the same mobile genetic elements [24]. A recent study demonstrated that river isolated *staphylococci* showed an extra chromosomal independent mechanism of resistance to multiple heavy metals and multiple antimicrobials [25]. This study also demonstrated plasmid-harbouring *Staphylococcus* isolates did not show any effect in their resistance profile to multiple antimicrobials and heavy metals despite the elimination of all plasmids [25]. These observations imply that the multi-drug and multi-metal resistance of river isolates of *Staphylococcus* is determined by chromosomal resistance.

In addition, other studies have reported that the expression of bacterial antimicrobial resistance systems were induced by heavy metals, for example, the transcription of multidrug efflux pump genes acrD and mdtABC in *Salmonella enterica*, were induced by a two-component signal transduction system BaeRS in response to

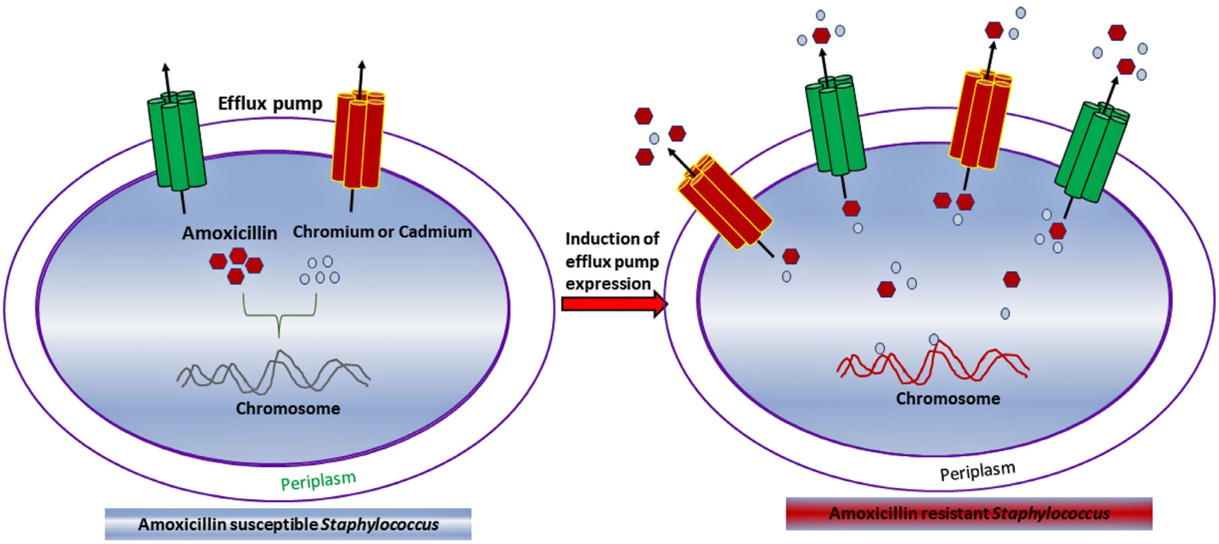

Fig. 7. Proposed model for the emergence of amoxicillin resistant *S. aureus* following chromium/cadmium and amoxicillin pre-exposure.

Table 4
Concentration and purity of RNA of *S. aureus* and the synthesized cDNA.

Condition	Sample	RNA concentration (ng/ µl)	cDNA concentration (ng/ µl)	Absorbance ratio (260nm/280nm)	Absorbance ratio (260nm/230nm)
48 hours pre-exposed *S. aureus*	Control	160.4	1500.1	1.82	2.15
	Cr^{6+}	167.0	1762.2	1.75	2.07
	Ax	126.0	1165.3	1.83	2.23
	Cr^{6+}+Ax	179.3	1352.3	1.81	2.18
	Cr^{6+}	167.0	1262.0	1.82	2.22
	Ax	360.1	1240.9	1.84	2.23
	Cr^{6+}+Ax	135.9	1219.2	1.83	2.21
	Ax	149.9	1297.5	1.84	2.24
	Cr^{6+}+Ax	134.2	1226.0	1.82	2.19
Unexposed *S. aureus*	Control	235.9	1279.0	1.82	2.21
	Cr^{6+}	240.3	1144.1	1.83	2.23
	Ax	284.0	1252.4	1.83	2.23
	Cr^{6+}+Ax	270.9	1127.1	1.81	2.21
	Cr^{6+}	236.8	1316.7	1.83	2.20
	Cr^{6+}+Ax	158.9	1074.5	1.82	2.22

copper or zinc resulting in enhanced antibiotic resistance [22]. Furthermore, chromate or copper modifies SoxS regulator and hence enhances the expression of multi-drug efflux pump AcrAB-TolC in *E. coli* (26). These proposed efflux pumps (e.g., AcrAB-TolC) are mostly responsible for conferring resistance against diverse antibiotics [26]. Recently, in a tissue cage infection model study demonstrated that in-complete or improper regimen of amoxicillin against *S. aureus* cause to induce amoxicillin resistance genes such as mecA, femA, femB and femX gene [23]. Another study has demonstrated that heavy metals with no nutritional benefits for microorganisms cause oxidative stress [27] that might trigger adaptations and bacterial growth in heavy metals and antimicrobial co-contamination settings. Presumably, high and chronic exposure to heavy metals and metals cause irreversible damage to bacterial DNA and the cell membrane, which later acts as an environmental selector for cellular defence [28]. This defence pathway could habituate bacteria to grow in presence of heavy metals, which may contribute to acquire resistance to heavy metals and antibiotics as a cross-resistance fashion. The recent study on bacterial resistance to heavy metals supported these earlier observations, where they showed that whether an intrinsic, natural, or a selective pressure-induced modification, bacteria acquire antimicrobial resistance through increases co- and/or cross-resistance pathways [29]. These observations and current study findings have strongly supported a mechanistic explanation (e.g., cross-resistance/regulation) behind the emergence of amoxicillin resistant *S. aureus* in the heavy metal and antibiotic co-contamination setting.

5. Conclusion

This study highlighted the co-exposure of chromium or cadmium and a low concentration of amoxicillin on the emergence of amoxicillin resistant *S. aureus*. This de novo emergence antibiotic resistant bacterium was validated through a culture-based technique. Further research is warranted to investigate the role of efflux pump in developing the resistance against amoxicillin in other bacterial population. Furthermore, studies should look at clinically relevant antimicrobials such as semi-synthetic penicillins used to treat *S. aureus*.

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Competing interests: The authors declare no competing interests.

Ethical approval: This research work is ethically approved. Ethical approval was given by the ethical review committee, Department of Biochemistry, University of Dhaka. The approval ID is BMBDU-ERC/EC/21/04.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2023.10.011.

References

[1] Zhang R, Eggleston K, Rotimi V, Zeckhauser RJ. Antibiotic resistance as a global threat: evidence from China, Kuwait and the United States. Global Health 2006;2:6.
[2] Baker-Austin C, Wright M, Stepanauskas R, McArthur J. Co-selection of antibiotic and metal resistance. Trends Microbiol 2006;14:176–82.
[3] Seiler C, Berendonk T. Heavy metal driven co-selection of antibiotic resistance in soil and water bodies impacted by agriculture and aquaculture. Front Microbiol 2012;3.
[4] Ahmad JU, Goni MA. Heavy metal contamination in water, soil, and vegetables of the industrial areas in Dhaka, Bangladesh. Environ Monit Assess 2010;166:347–57.
[5] Gupta N, Khan DK, Santra SC. An assessment of heavy metal contamination in vegetables grown in wastewater-irrigated areas of Titagarh, West Bengal, India. Bull Environ Contam Toxicol 2008;80:115–18.
[6] Martínez JL, Rojo F. Metabolic regulation of antibiotic resistance. FEMS Microbiol Rev 2011;35:768–89.
[7] Zhu Y-G, Johnson TA, Su J-Q, Qiao M, Guo G-X, Stedtfeld RD, et al. Diverse and abundant antibiotic resistance genes in Chinese swine farms. Proc Natl Acad Sci USA 2013;110:3435–40.
[8] Looft T, Johnson TA, Allen HK, Bayles DO, Alt DP, Stedtfeld RD, et al. In-feed antibiotic effects on the swine intestinal microbiome. Proc Natl Acad Sci USA 2012;109:1691–6.
[9] Chen S, Li X, Sun G, Zhang Y, Su J, Ye J. Heavy metal induced antibiotic resistance in bacterium LSJC7. Int J Mol Sci 2015;16:23390–404.
[10] Heuer H, Schmitt H, Smalla K. Antibiotic resistance gene spread due to manure application on agricultural fields. Curr Opin Microbiol 2011;14:236–43.
[11] Khan S, Cao Q, Zheng YM, Huang YZ, Zhu YG. Health risks of heavy metals in contaminated soils and food crops irrigated with wastewater in Beijing. China. Environ Pollut 2008;152:686–92.
[12] Knapp CW, McCluskey SM, Singh BK, Campbell CD, Hudson G, Graham DW. Antibiotic resistance gene abundances correlate with metal and geochemical conditions in archived Scottish soils. PLoS One 2011;6:e27300.
[13] Wiegand I, Hilpert K, Hancock REW. Agar and broth dilution methods to determine the minimal inhibitory concentration (MIC) of antimicrobial substances. Nat Protoc 2008;3:163–75.
[14] Bera A, Biswas R, Herbert S, Kulauzovic E, Weidenmaier C, Peschel A, et al. Influence of wall teichoic acid on lysozyme resistance in Staphylococcus aureus. J Bacteriol 2007;189:280–3.
[15] Atshan SS, Shamsudin MN, Lung LTT, Ling KH, Sekawi Z, Pei CP, et al. Improved method for the isolation of RNA from bacteria refractory to disruption, including S. aureus producing biofilm. Gene 2012;494:219–24.
[16] Huet AA, Raygada JL, Mendiratta K, Seo SM, Kaatz GW. Multidrug efflux pump overexpression in Staphylococcus aureus after single and multiple in vitro exposures to biocides and dyes. Microbiology 2008;154(Pt 10):3144–53.
[17] Livak KJ, Schmittgen TD. Analysis of relative gene expression data using real-time quantitative PCR and the 2(-Delta Delta C(T)) method. Methods 2001;25:402–8.
[18] Islam LN, Rahman A, Mahmud Z, Nabi AHMN, Hossain M, Mohasin M. Assessment of physicochemical and biochemical qualities of tannery effluents of Hazaribagh, Dhaka, and comparison with non-tannery water samples. Int J Environ 2015;4:68–81.
[19] Xue H, Wu Z, Li L, Li F, Wang Y, Zhao X. Coexistence of heavy metal and antibiotic resistance within a novel composite staphylococcal cassette chromosome in a Staphylococcus haemolyticus isolate from bovine mastitis milk. Antimicrob Agents Chemother 2015;59:5788–92.
[20] Nguyen CC, Hugie CN, Kile ML, et al. Association between heavy metals and antibiotic-resistant human pathogens in environmental reservoirs: a review. Front Environ Sci Eng 2019;46.
[21] Pal C, Bengtsson-Palme J, Kristiansson E, Larsson DG. Co-occurrence of resistance genes to antibiotics, biocides and metals reveals novel insights into their co-selection potential. BMC Genom 2015;16:964.
[22] Nishino K, Nikaido E, Yamaguchi A. Regulation of multidrug efflux systems involved in multidrug and metal resistance of Salmonella enterica serovar Typhimurium. J Bacteriol 2007;189:9066–75.
[23] Yao Q, Gao L, Xu T, Chen Y, Yang X, Han M, et al. Amoxicillin administration regimen and resistance mechanisms. Front Microbiol 2019;10:1638.
[24] Ye J, Rensing C, Su J, Zhu YG. From chemical mixtures to antibiotic resistance. J Environ Sci (China) 2017;62:138–44.
[25] Yilmaz F, Orman N, Serim G, Kochan C, Ergene A, Icgen B. Surface water-borne multidrug and heavy metal-resistant Staphylococcus isolates characterized by 16S rDNA sequencing. Bull Environ Contam Toxicol 2013;91:697–703.
[26] Harrison JJ, Tremaroli V, Stan MA, Chan CS, Vacchi-Suzzi C, Heyne BJ, et al. Chromosomal antioxidant genes have metal ion-specific roles as determinants of bacterial metal tolerance. Environ Microbiol 2009;11:2491–509.
[27] Jaishankar M, Tseten T, Anbalagan N, Mathew BB, Beeregowda KN. Toxicity, mechanism and health effects of some heavy metals. Interdiscip Toxicol 2014;7(2):60–72.
[28] Safari Sinegani AA, Younessi N. Antibiotic resistance of bacteria isolated from heavy metal-polluted soils with different land uses. J Glob Antimicrob Resist 2017;10:247–55.
[29] Gorovtsov AV, Sazykin IS, Sazykina MA. The influence of heavy metals, polyaromatic hydrocarbons, and polychlorinated biphenyls pollution on the development of antibiotic resistance in soils. Environ Sci Pollut Res Int 2018;25:9283–92.